Horizontal and vertical dilations

1

a 2π

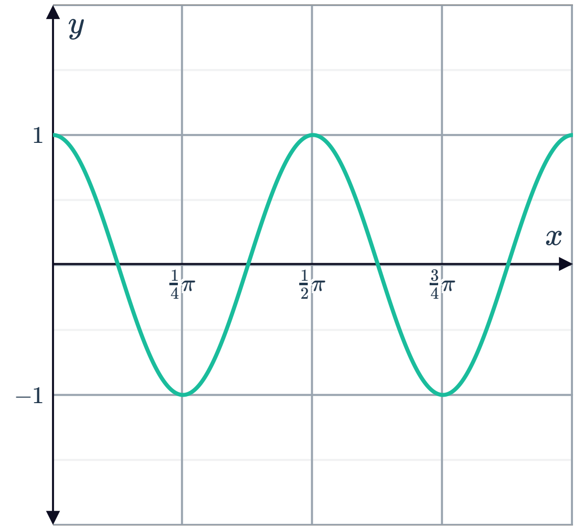
b

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$	$\frac{5\pi}{8}$	$\frac{3\pi}{4}$	$\frac{7\pi}{8}$	π
$g(x)$	1	0	-1	0	1	0	-1	0	1

c $\frac{1}{2}\pi$

d Horizontal dilation by a factor of $\frac{1}{4}$

e

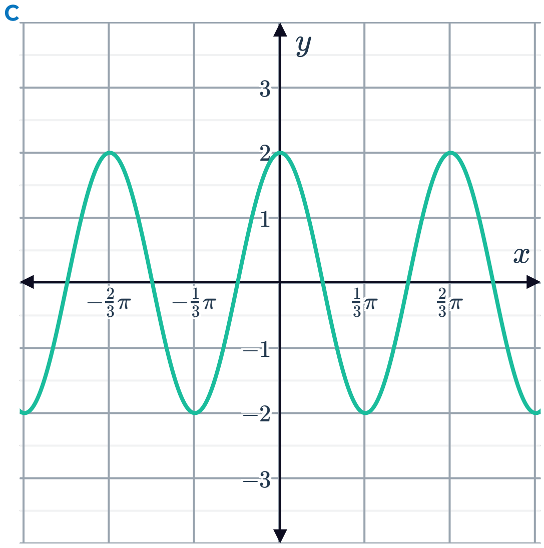


2

a 2

b $\frac{2\pi}{3}$

c



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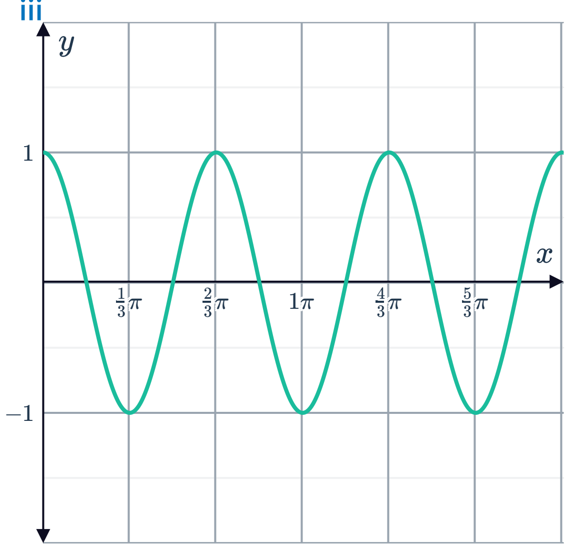
3

a

i $A = 1$

ii $P = \frac{2\pi}{3}$

iii

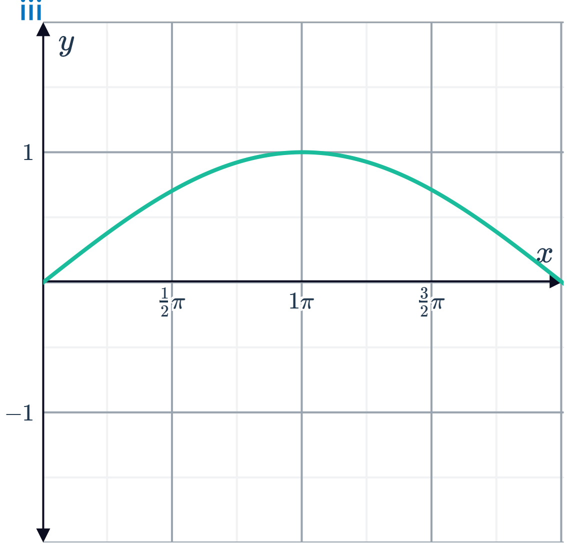


b

i $A = 1$

ii $P = 4\pi$

iii

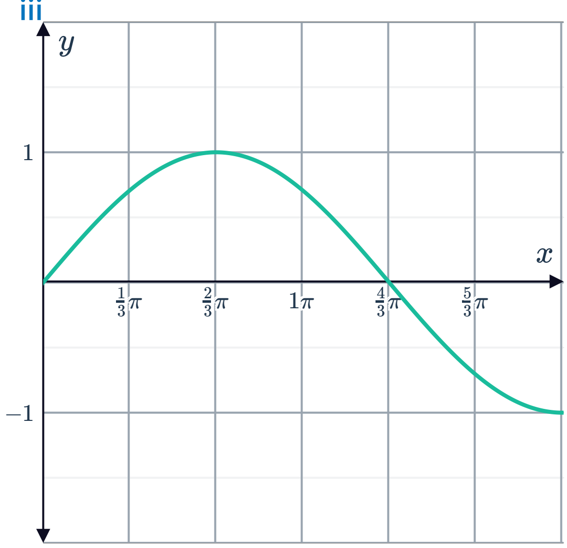


c

i $A = 1$

ii $P = \frac{8\pi}{3}$

iii



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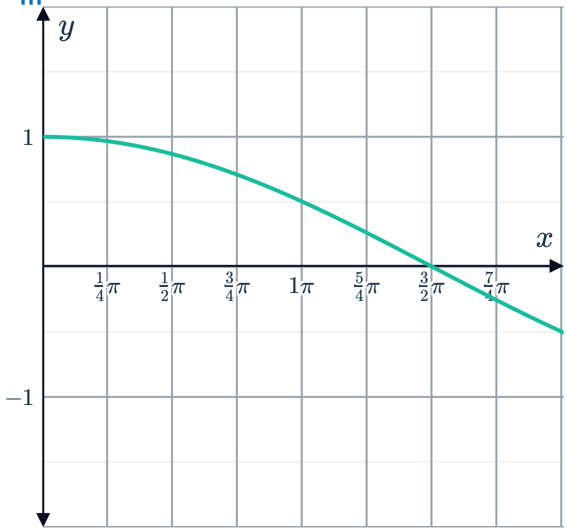
d

i $A = 1$

ii $P = 6\pi$

iii

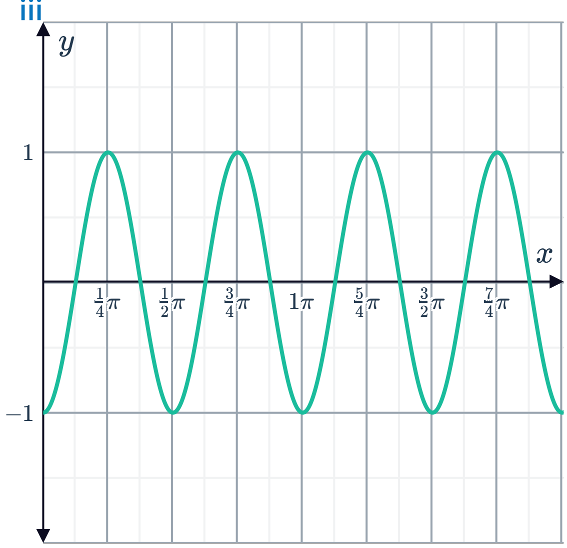


e

i $A = 1$

ii $P = \dfrac{\pi}{2}$

iii

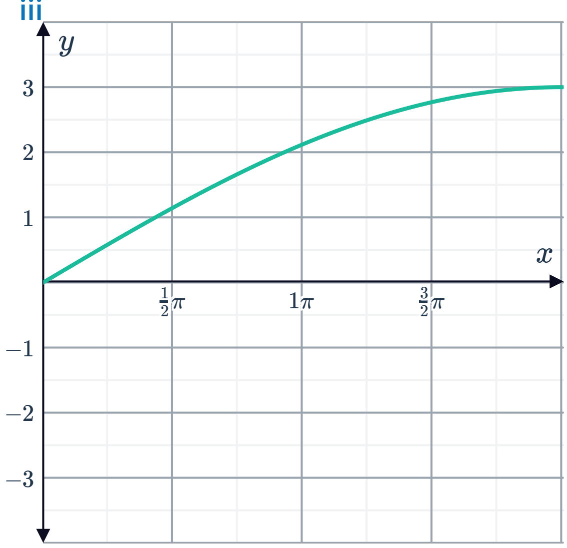


f

i $A = 3$

ii $P = 8\pi$

iii



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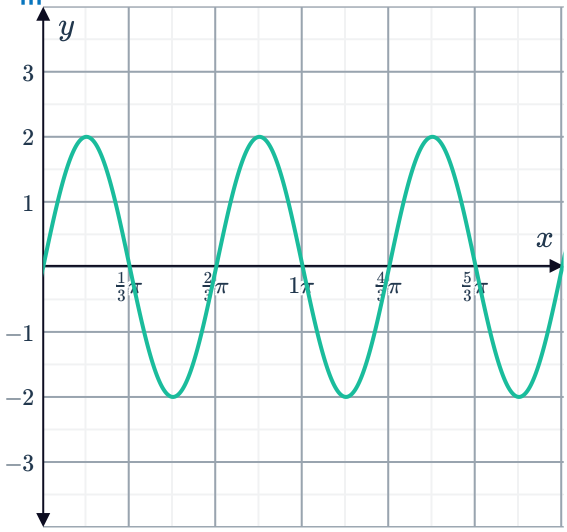
g

i $A = 2$

ii $P = \dfrac{2\pi}{3}$

iii

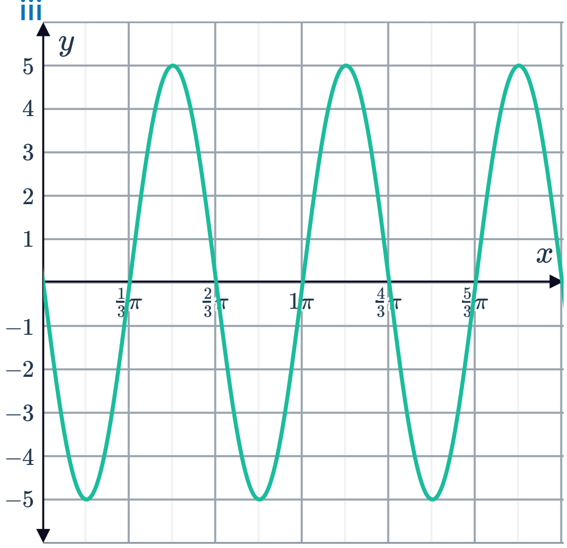


h

i $A = 5$

ii $P = \dfrac{2\pi}{3}$

iii

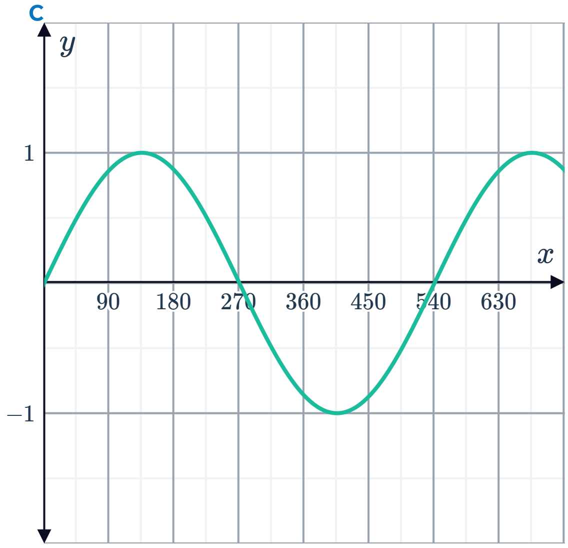


4

a 1

b 540°

c



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a

x	0	2π	4π	6π	8π
$\sin\left(\frac{x}{4}\right)$	0	1	0	-1	0

b 8π

6

a

Function	Period
$\tan x$	π
$\tan 2x$	$\frac{\pi}{2}$
$\tan 3x$	$\frac{\pi}{3}$
$\tan 4x$	$\frac{\pi}{4}$

b $\frac{\pi}{b}$

c Shorter

7

a $\frac{2\pi}{3}$

b 1

c $\frac{\pi}{2}$

d 4π

e 2

f $\frac{4\pi}{3}$

g $\frac{3\pi}{2}$

h 2

8

a Horizontal dilation by a factor of $\frac{1}{4}$.

b 4

9

a

i 0

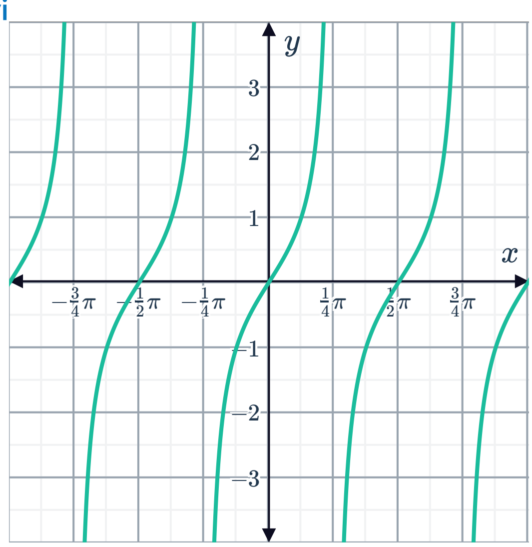
ii $\frac{\pi}{2}$

iii $\frac{\pi}{2}$

iv $x = \frac{\pi}{4}$

v $x = -\frac{\pi}{4}$

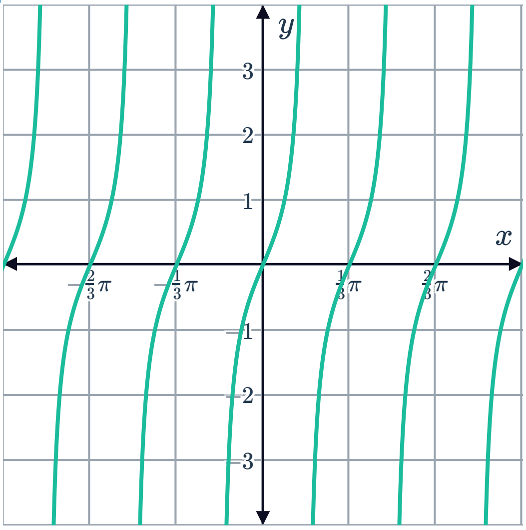
vi



b

i 0 **ii** $\frac{\pi}{3}$ **iii** $\frac{\pi}{3}$ **iv** $x=\frac{\pi}{6}$

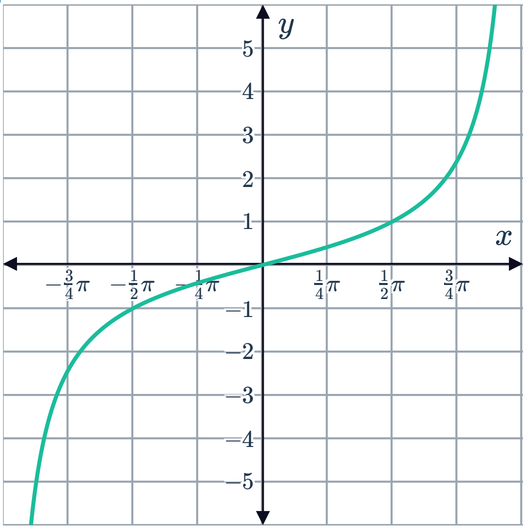
v $x=-\frac{\pi}{6}$ **vi**



c

i 0 **ii** 2π **iii** 2π **iv** $x=\pi$

v $x=-\pi$ **vi**



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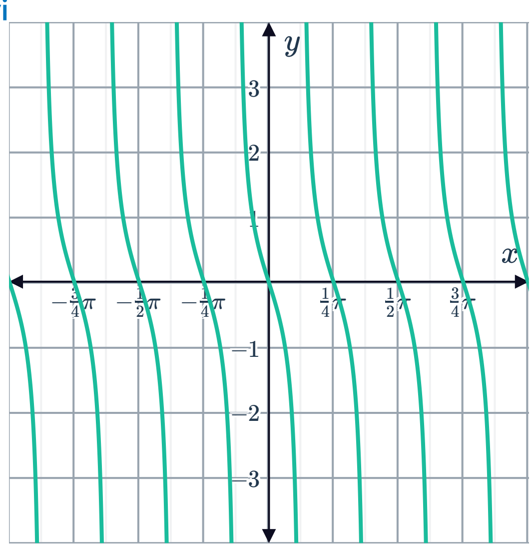
d

i 0 **ii** $\frac{\pi}{4}$ **iii** $\frac{\pi}{4}$ **iv** $x = \frac{\pi}{8}$

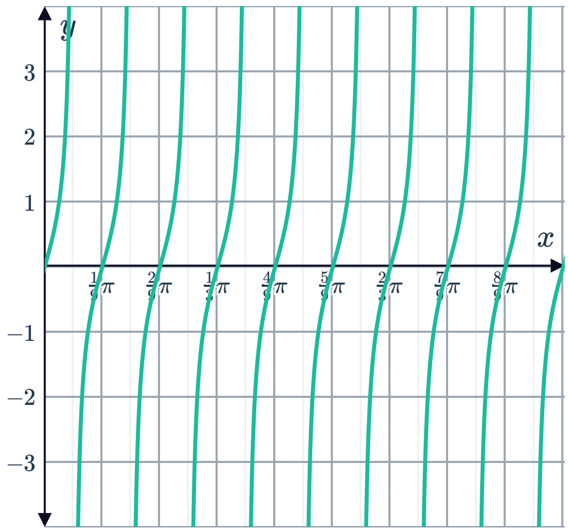
v $x = -\frac{\pi}{8}$ **vi**



10

a $\frac{\pi}{9}$

b



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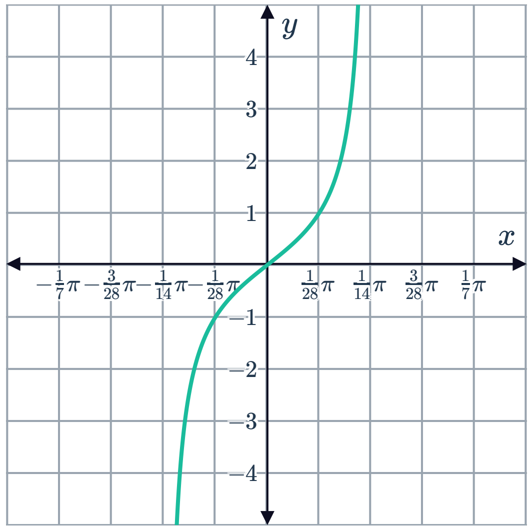
11

a

x	$-\frac{\pi}{28}$	0	$\frac{\pi}{28}$	$\frac{3\pi}{28}$	$\frac{\pi}{7}$	$\frac{5\pi}{28}$
y	-1	0	1	-1	0	1

b



12

a $\frac{53\pi}{24}$ **b** $\frac{\pi}{12}$ **c** $y = \tan 12x$

13 β is larger as $g(x)$ has a shorter period.

14

a

x	0	$\frac{3\pi}{2}$	3π	$\frac{9\pi}{2}$	6π
$\tan\left(\frac{x}{6}\right)$	0	1	undefined	-1	0

b 6π

d $x = 3\pi,\ x = 9\pi,\ x = 15\pi$

e -3π

Phase shifts

15

a 1 **b** -1

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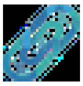
16

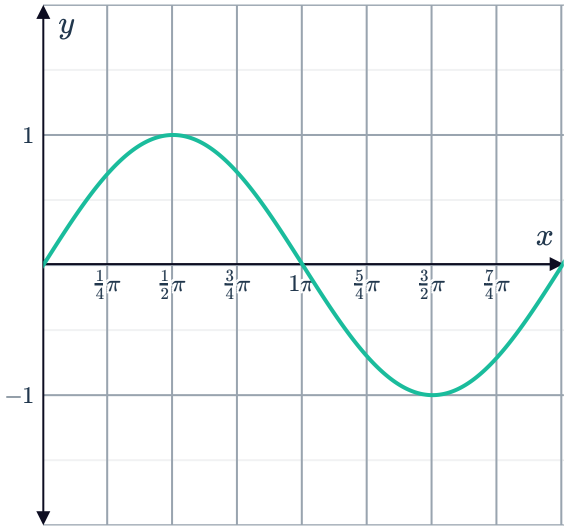
a

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$	1	0	-1	0	1
$g(x)$	0	1	0	-1	0



b Translated $\frac{\pi}{2}$ units to the right.

c



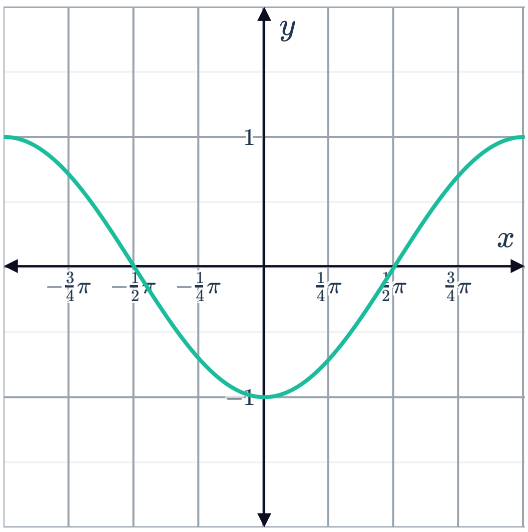
17

a

i 1

ii Translated $\frac{\pi}{2}$ units to the right.

iii



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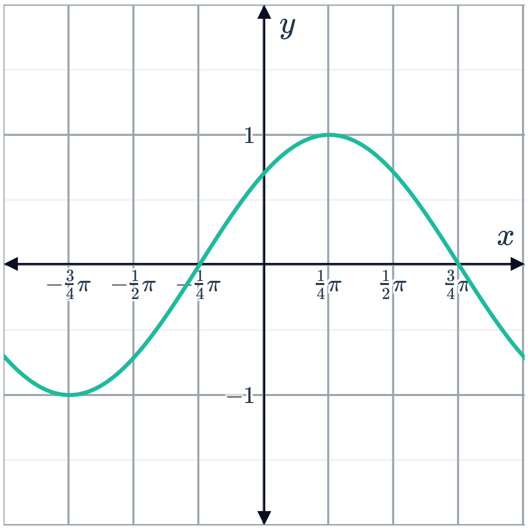
b

i 1

ii Translated $\frac{\pi}{4}$ units to the left.

iii



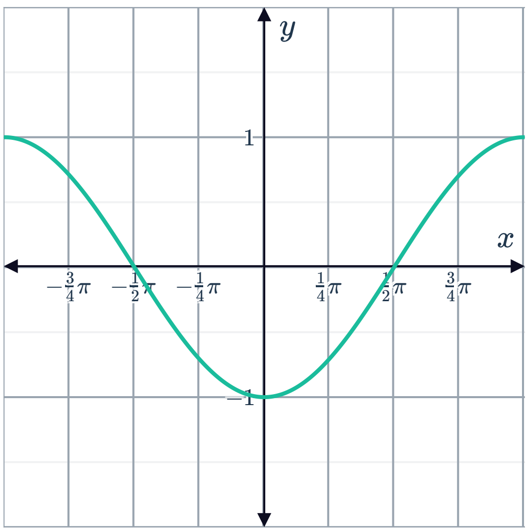
18

a

i 1

ii Translated π units to the left.

iii



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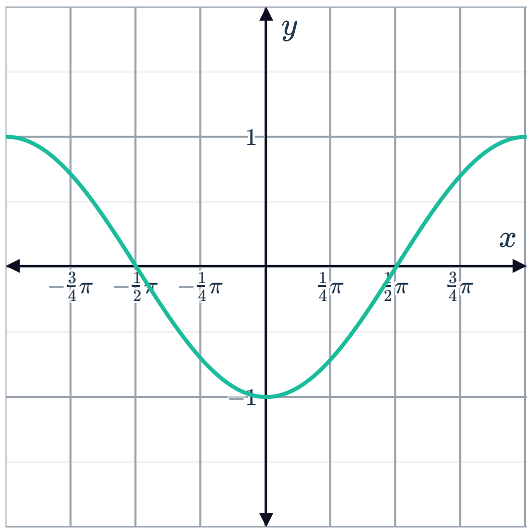
b

i 1

ii Translated π units to the right.

iii

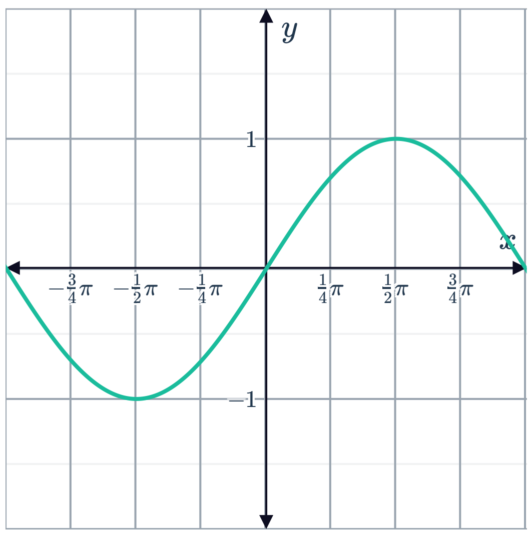


c

i 1

ii Translated $\dfrac{3\pi}{2}$ units to the left.

iii



19 $c = -\dfrac{1}{2}\pi,\ \dfrac{3}{2}\pi$

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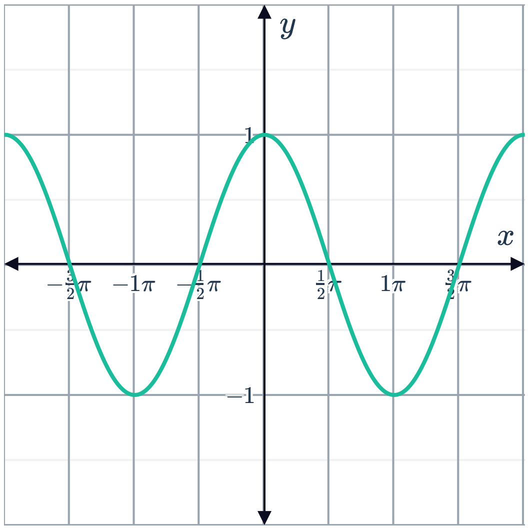
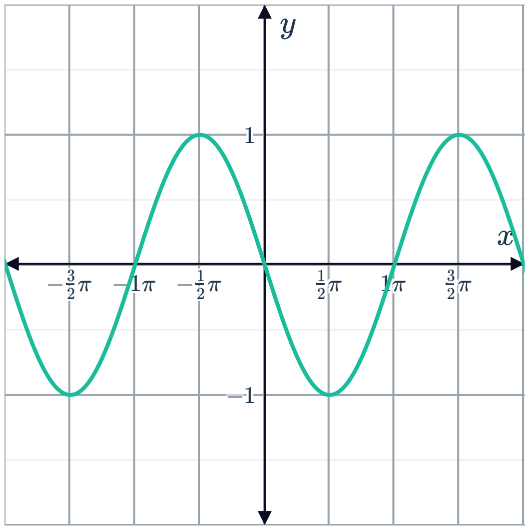
20

a



b



21 True

22

a $\frac{\pi}{2}$ units to the left.

b $\frac{\pi}{5}$ units to the right.

c 1.5 units to the right.

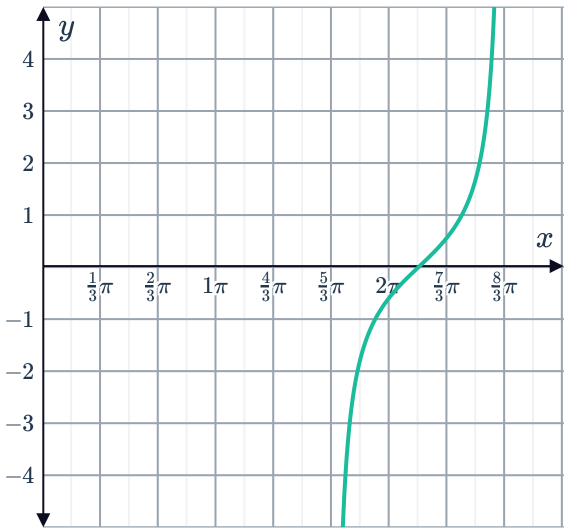
d $\frac{3\pi}{4}$ units to the left.

23

a $x = \frac{2\pi}{3}, x = \frac{5\pi}{3}, x = \frac{8\pi}{3}$

b $x = \frac{7\pi}{6}, \frac{13\pi}{6}$

c



24

a

i There is no y-intercept.

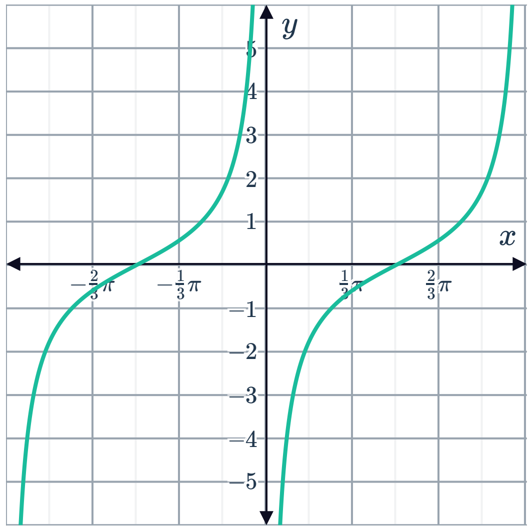
ii π

iii π

iv $x = \pi$

v $x = 0$

vi



b

i $-\dfrac{1}{\sqrt{3}}$

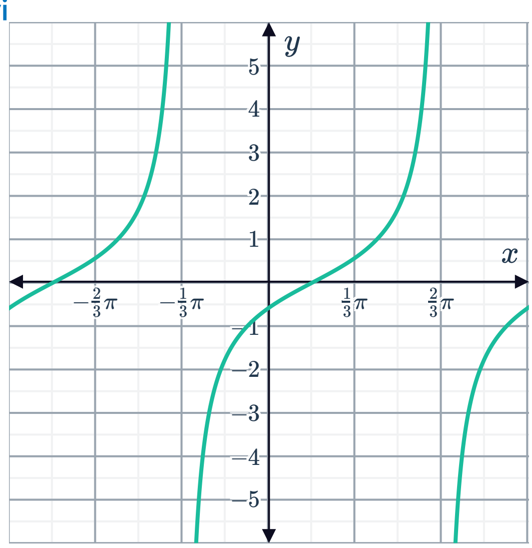
ii π

iii π

iv $x = \dfrac{2\pi}{3}$

v $x = -\dfrac{\pi}{3}$

vi



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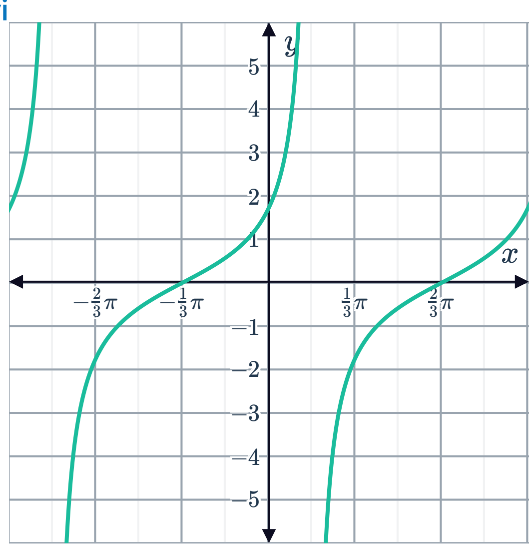
c

i $\sqrt{3}$ **ii** π **iii** π **iv** $x = \frac{\pi}{6}$

v $x = -\frac{5\pi}{6}$ **vi**



25

a $\frac{\pi}{7}$ units to the right

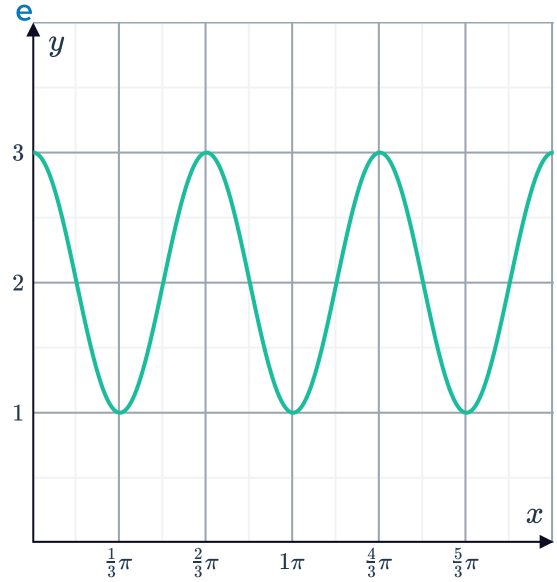
b $x = \frac{9\pi}{14}, x = \frac{23\pi}{14}, x = \frac{37\pi}{14}, x = \frac{51\pi}{14}$

Combined translations and dilations

26

a $\frac{2\pi}{3}$ **b** 1 **c** 3 **d** 1

e



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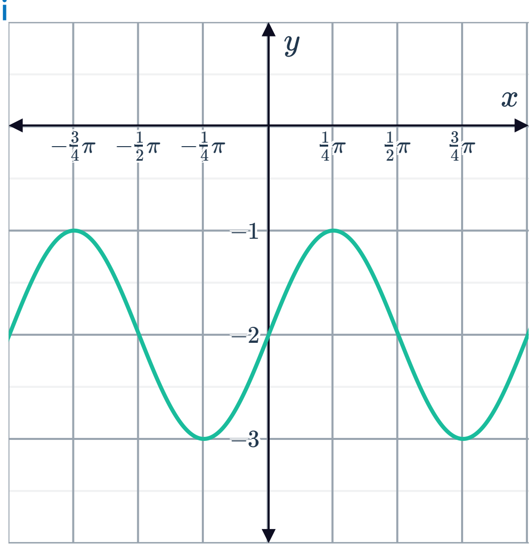
27

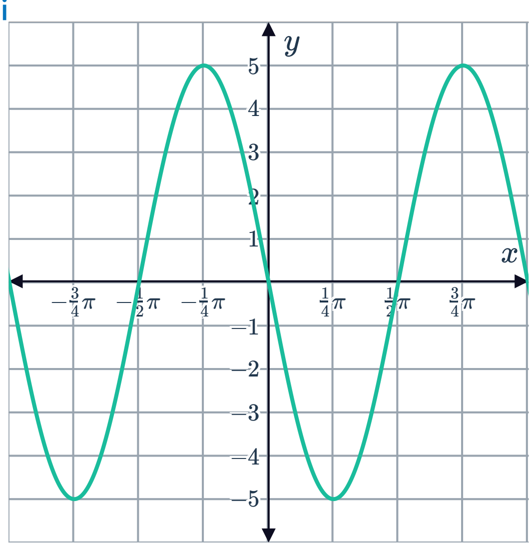
a

i $(-\infty, \infty)$ **ii** $[-3, -1]$ **iii**



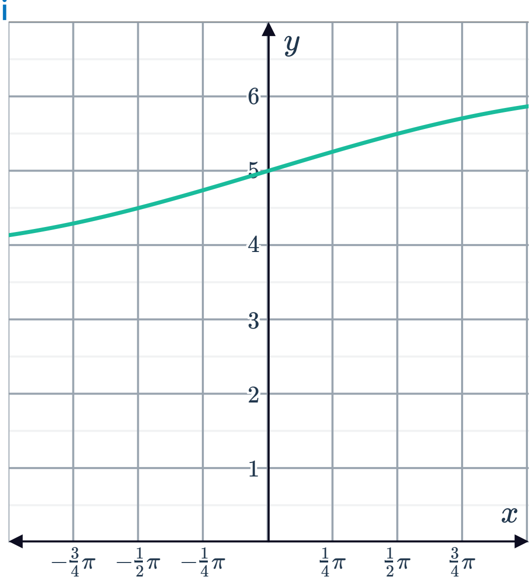
b

i $(-\infty, \infty)$ **ii** $[-5, 5]$ **iii**



c

i $(-\infty, \infty)$ **ii** $[4, 6]$ **iii**



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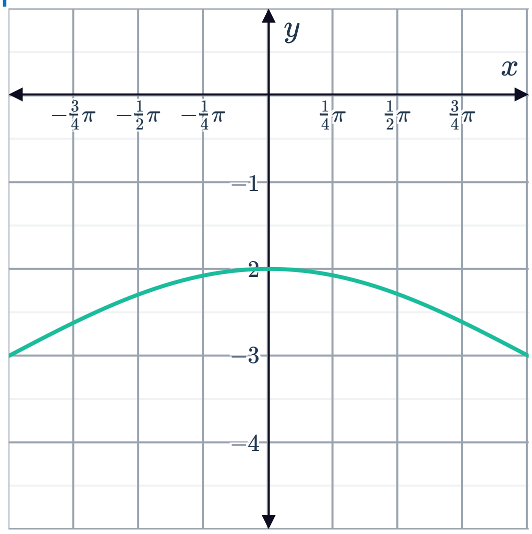
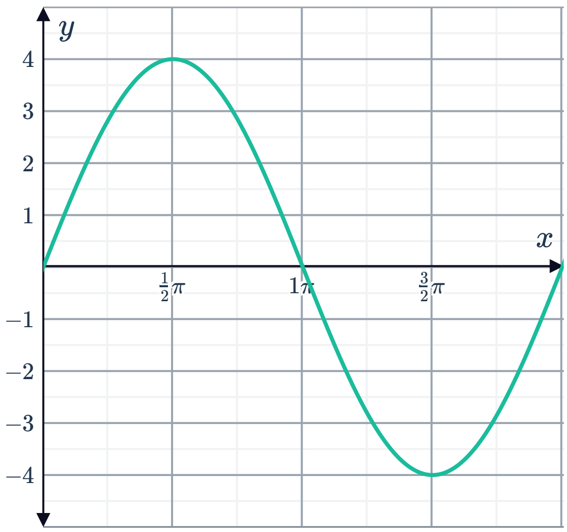
d

i $(-\infty, \infty)$ **ii** $[-4, -2]$ **iii**

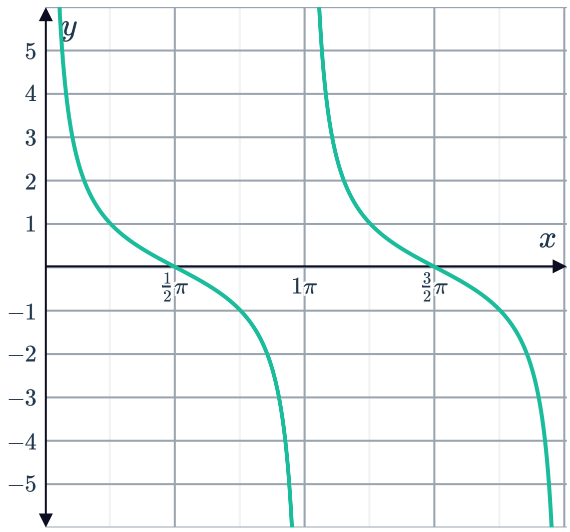


28

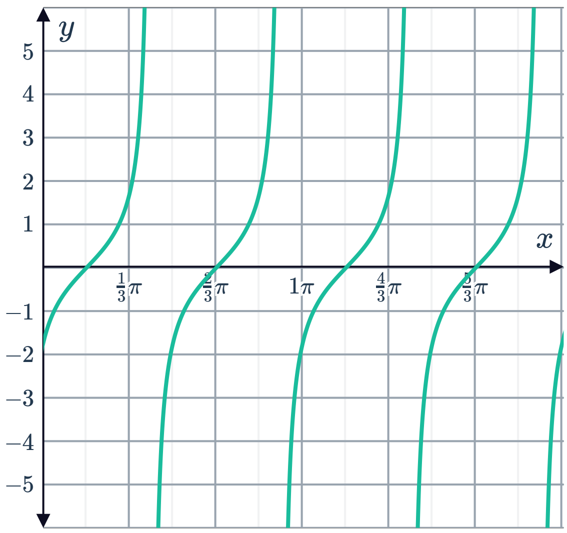
a



b



c



29 Dilated vertically by a factor of 3 and a phase shift of π to the right

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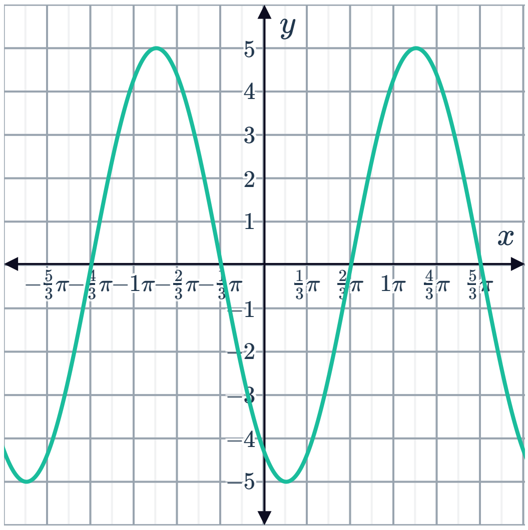
30

a Reflected about the x-axis, then dilated vertically by a factor of 5, followed by a phase shift of $\frac{\pi}{3}$ units to the left.

b



31 $y = -\cos\left(x + \frac{\pi}{6}\right) + 5$

32

a $y = 3\sin 2x$ **b** $y = -\sin 2x$ **c** $y = 2\cos 3x$ **d** $y = -4\cos\left(\frac{x}{2}\right)$

33

a 3

b $\frac{\pi}{3}$

c Translated 3 units downwards and a phase shift of $\frac{\pi}{3}$ to the right.

34 Translated vertically by 2 units upwards and a phase shift of $\frac{\pi}{4}$ units to the right.

35

a Yes **b** No **c** No **d** Yes

e No

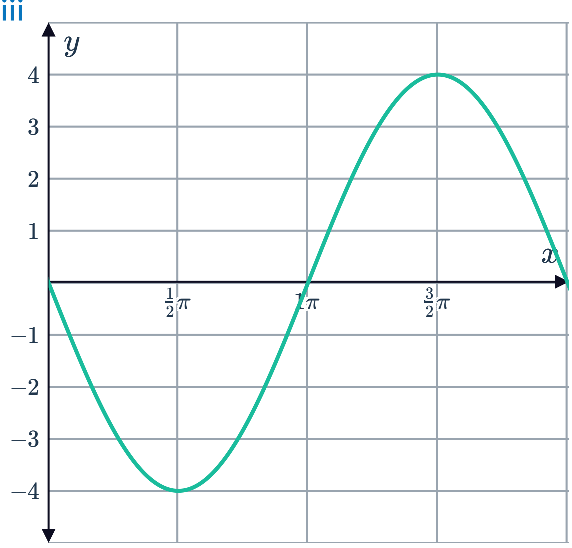
36

a

i 4

ii Translated π units to the right.

iii

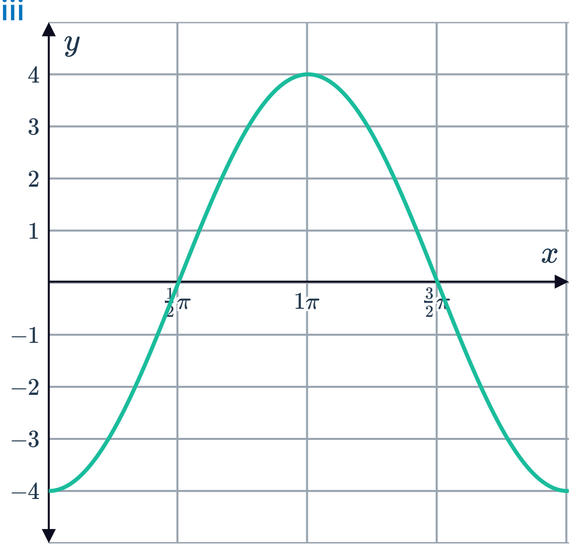


b

i 4

ii Translated $\frac{\pi}{2}$ units to the left.

iii



37

a Horizontal dilation by a factor of $\frac{1}{2}$.

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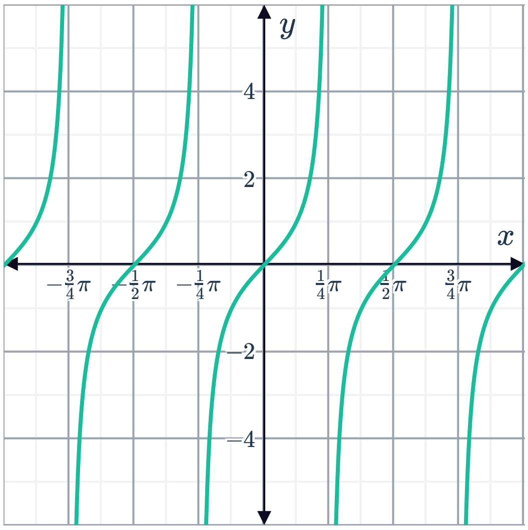
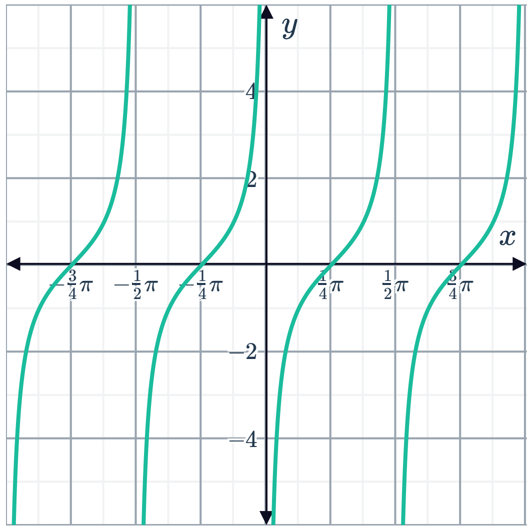
b

i



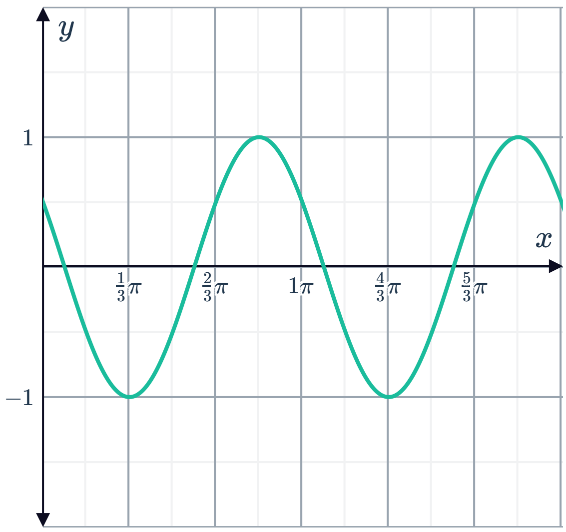
ii



38

a Horizontal dilation by a factor of $\frac{1}{2}$ and then a phase shift of $\frac{\pi}{6}$ to the left.

b



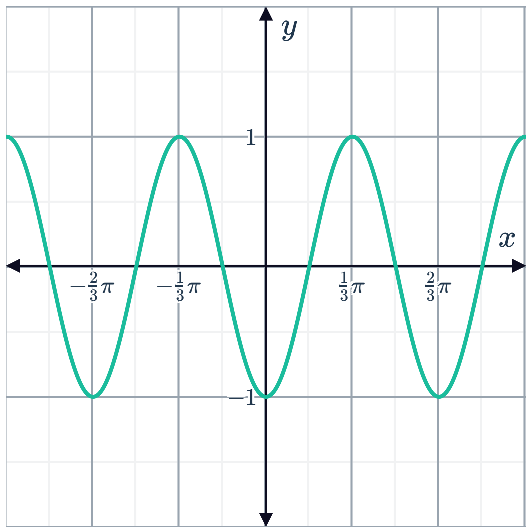
39 A horizontal dilation of scale factor $\frac{1}{2}$ followed by a horizontal translation of $\frac{\pi}{4}$ units to the right.

40

a $\frac{2\pi}{3}$

b Horizontal dilation by a factor of $\frac{1}{3}$ followed by a phase shift of $\frac{\pi}{3}$ units to the left.

c

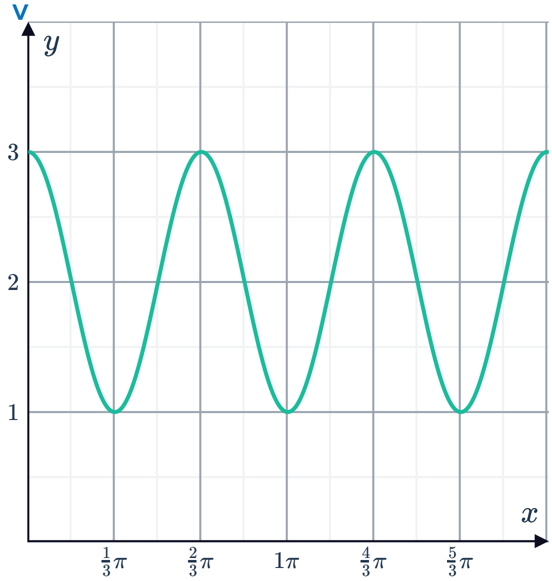


41

a

i $\frac{2\pi}{3}$ **ii** 1 **iii** 3 **iv** 1

v



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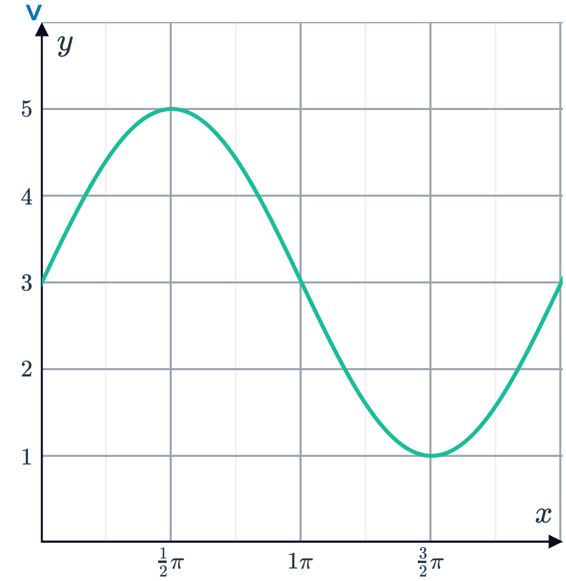
b

i 2π **ii** 2 **iii** 5 **iv** 1

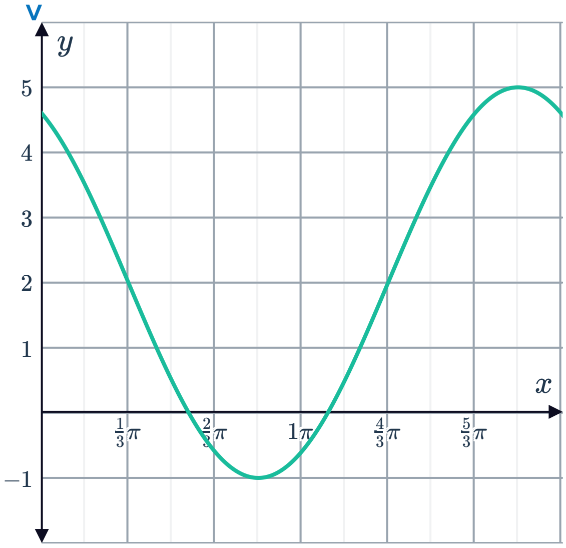
v



c

i 2π **ii** 3 **iii** 5 **iv** -1

v



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42

a 5π

b

- Reflected about the x-axis.
- Vertically dilated by a factor of 4.
- Horizontally dilated by a factor of 5.
- Horizontally translated by $\frac{\pi}{4}$ units to the left.

c $(-\infty, \infty)$

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